ProMaxx Advanced Fuels™

COAL TO DIESEL

Advanced Fuel System

coal

diesel

electricity

Imagine independently producing high value diesel fuel on site, using your own coal feedstock at a lower cost, with lower fuel emissions and generating higher machine operating efficiency.

ProMaxx Advanced Fuel systems makes independent production of premium grade Fischer-Tropsch fuels possible.

ABOUT
FISCHER - TROPSCH

Fischer-Tropsch (FT) technology has existed for nearly a hundred years. The FT process was developed in the 1920s at the Kaiser-Wilhelm-Institute for Chemistry by two German scientists: Franz Fischer and Hans Tropsch. Lacking petroleum reserves, Germany turned to FT technology during World War II in order to convert its abundant coal reserves into usable fuel. This reliance on FT synthetic fuel allowed Germany to become less dependent on other countries for fuel during the war.

After World War II, nations and companies around the world have been steadily turning to FT technology. Virtually every major oil company is successfully utilizing, or in the process of building, large scale FT facilities. In 1993 Shell built its first FT facility in Malaysia and has since built several new FT projects around the world. BP has also built an FT trial facility in Alaska.

Like the major oil companies, ProMaxx Advanced Fuels is at the forefront of refining FT technology. However, unlike the major oil companies ProMaxx Advanced Fuels can deploy small-scale units, in a much shorter timeframe and at a fraction of the cost.

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ABOUT
PROMAXX ADVANCED FUELS

As the nation's fossil fuel resources gradually deplete, companies and governments are trying to find smart ways to maximize the value of natural resources and reduce reliance on expensive foreign fuel imports. With coal prices down in recent years companies are also looking for ways to cut costs, increase efficiency and maximize margins.

ProMaxx Advanced Fuels systems, manufactured in the United States, offers companies a profitable solution to reduce operational costs while also replacing logistical uncertainty with the security of producing your own high quality FT diesel on site.

ProMaxx Advanced Fuel systems deliver site-based, modular solutions for independent production of high quality FT diesel to maximize revenue while reducing environmental impacts and logistical risks associated with fuel delivery in remote areas.

INTRODUCTION
TO THE ADVANCED FUEL SYSTEM

ProMaxx Advanced Fuels has developed and deployed systems to enable year-round, independent diesel fuel production using coal from your own mine.

ProMaxx Advanced Fuel pyrolysis systems convert coal into premium FT diesel.

With this advanced system comes several key benefits:

- Cost effective production of premium liquid fuel
- On-site installation and operation
- Independent year-round supply of FT diesel
- Increased operational efficiency for machinery
- Environmentally friendly

BENEFITS :
PREMIUM SYNTHETIC FT DIESEL

Our licensed and patented system takes in all types of coal to produce a high value diesel fuel. FT technology produces a high cetane (70+), ultra-clean diesel fuel with zero sulphur, aromatics or heavy metals and demonstrates a high operating efficiency.

Studies show FT fuels with the attributes found in ProMaxx Advanced Fuels diesel, reduces emissions and helps optimize diesel engine performance. These characteristics also make FT diesel desirable as it provides both a performance boost and an emission reduction for diesel consumers.

High cetane diesel can also be used as an additive by fuel blenders and is therefore a very desirable product that can be sold at a premium compared to regular diesel.

Comparison Table	Specific Gravity	Calorific Value (BTU)	Cetane Number
Diesel	0.83	9,240	45
FT Diesel	0.88	10,784	70+
Water	1.00	NA	NA

cost comparison diesel or benefit chart

Promaxx 50 TPD
Coal to FT Diesel System
in Australia

BENEFITS :
INDEPENDENT FUEL PRODUCTION

Through the deployment, installation and operation of cutting edge technologies such as ProMaxx Advanced Fuel's portable and modular systems, companies can reduce the cost of fuel supply and transport to the operating site, leading to increased efficiency and greater margins for the core business.

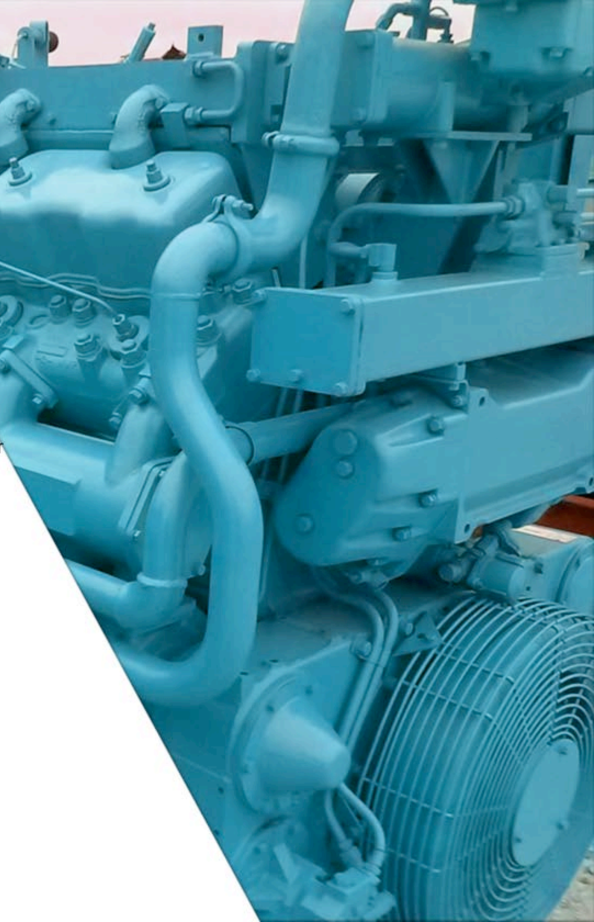
The modular nature of the systems permit companies to add additional units over time to increase on-site FT diesel production to match operational fuel requirements.

BENEFITS :
HIGHER OPERATING EFFICIENCY FOR RUNNING OF MACHINERY

With this superior product comes increased operating efficiency for machinery. Similar to gasoline, a higher cetane number results in increased engine performance. Compared to regular diesel, FT-diesel has a superior cetane level (i.e. 70+ compared to 45+) translating into increased power, torque and mileage for diesel engines.

As well as improved engine efficiency there are corresponding reductions in CO_2 and NO_x emissions. FT diesel contains zero Sulphur*, which means a substantial reduction in particulate emissions.

** Coal feedstock with greater than 1.5% sulphur content must be treated through a third party scrubber prior to use.*



BENEFITS :
ENVIRONMENTALLY FRIENDLY

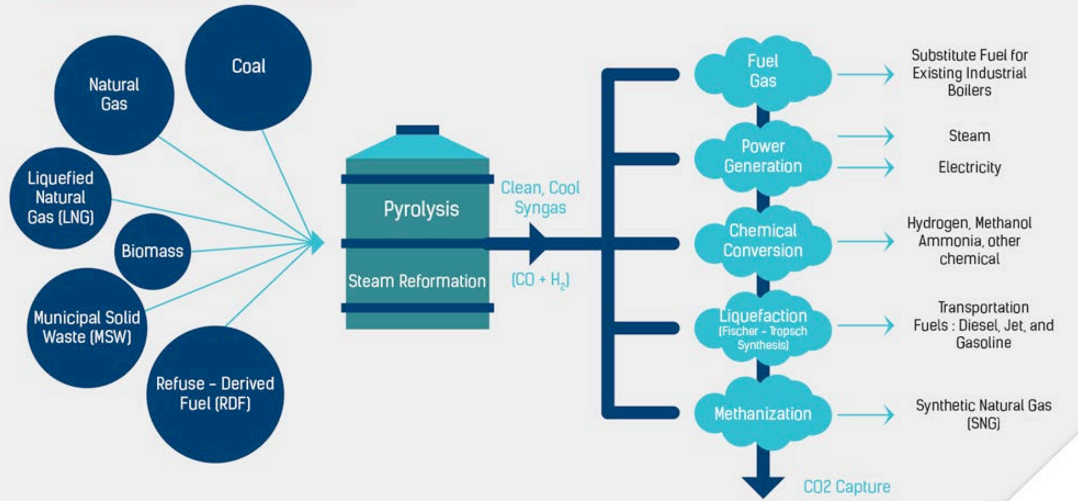
ProMaxx Advanced Fuel produces more power with fewer emissions. FT technology converts coal, natural gas, and low-value refinery products into a high-value, clean-burning fuel. There are no harmful or hazardous emissions produced in ProMaxx pyrolysis systems.

FT fuel is colorless, odor-less, and low in toxicity and has been endorsed by the US Environmental Protection Agency as an alternative clean fuel. It is interchangeable with conventional diesel fuels and can be blended with diesel at any ratio with no modification. FT fuels offer important emissions benefits compared with regular diesel, reducing nitrogen oxide, carbon monoxide, and particulate matter.

Regular Diesel

FT Diesel

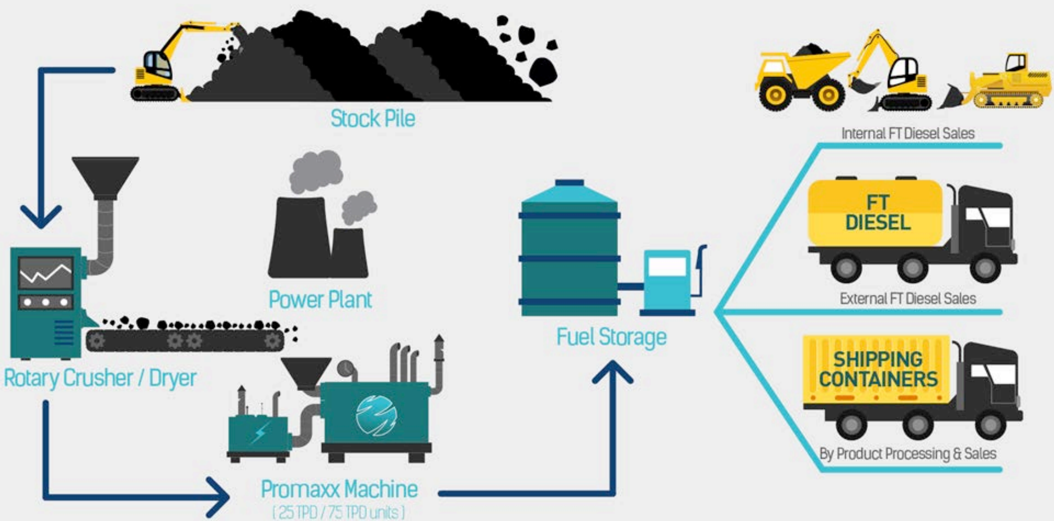
SYNGAS APPLICATION



HOW IT WORKS :

ProMaxx Advanced Fuel systems convert coal, natural gas, liquid gas and other hydrocarbon feed stocks in to FT diesel and other customizable synthetic fuels using pyrolysis and/or steam reformation to produce Syngas followed by FT liquefaction and compression.

COAL TO DIESEL PROCESS

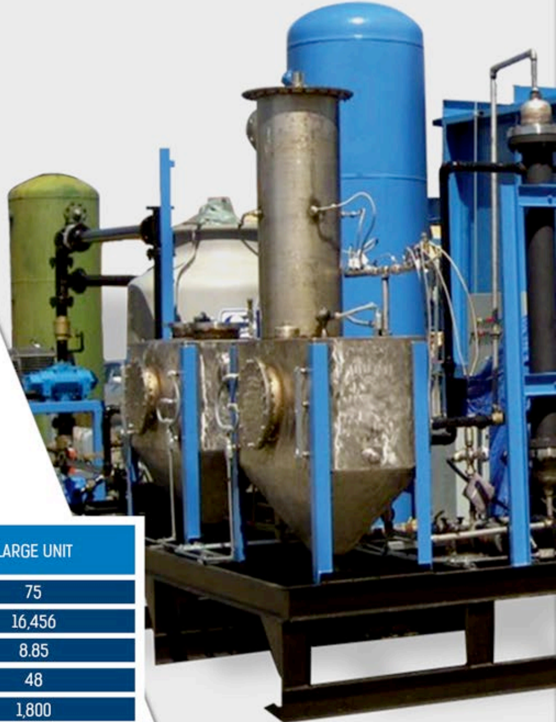


SYSTEM SPECIFICATION

ProMaxx custom builds advanced fuel systems to fit Client needs. At present our most popular models include the 25 and 75 ton input per day systems. The total number of modules deployed on site is simply scaled to meet Client fuel production requirements. FT diesel output is dependent on the calorific value and chemical composition of the coal feed stock. Machine conversion efficiency is typically in the range of 19-32%. The higher the calorific value of the coal feedstock, the greater the volume of FT diesel produced.

Coal with calorific value greater than 3,000 kcal/kg can be fed into ProMaxx systems, however low rank coal should be upgraded through a simple drying process to maximize FT diesel production. As the dried coal is used immediately after drying there is little or no re-absorption of water and therefore no briquetting or coating of the dried coal is required.

FT diesel production figures for both the 25 and 75 TPD systems are outlined below. These figures are based on 4,000 kcal/kg coal (as received), upgraded through drying to 7% moisture



MODEL	SMALL UNIT	LARGE UNIT
Coal Consumption (US Ton / Day)	25	75
FT Diesel Production (Liters / Day)	5,485	16,456
Carbon Black Production (Tonnes / Day)	2.95	8.85
Man Hours (per day)	48	48
Electricity (KWhr)	750	1,800
Electrical Service	480V / 1000A	480V / 1000A
System Footprint (meters)	W : 15 L : 22 H : 7	W : 15 L : 38 H : 7

CLIENT SALES PROCESS

ProMaxx Advanced Fuels is committed to helping customers produce energy commodities at the best cost-to-transformation ratio and ensuring that the projects placed in their customer's portfolio are commercially viable, economical and profitable. We work with Clients through a 10-step process to achieve this commitment:

1. Client presentation & needs assessment.
2. Signing of Non Disclosure and Non Circumvention Agreement.
   Completion of Request for Quotation (RFQ)
4. Delivery of a financial model based on Client specs.
5. Client decision to proceed with a sale.
6. ProMaxx factory visit and analysis of Client's coal feedstock.
7. Contract execution and down payment.
8. ProMaxx system manufacturing over 10-14 month period.
9. Client shipping to destination country.
10. ProMaxx on-site system commissioning and training.

Under Step 6, Customers are encouraged to bring an FT expert with them to view operation of a scaled-down version of the Coal to FT diesel system. ProMaxx professionals will then work with your engineer to process a 200 kg representative sample of your coal in to FT diesel and calculate FT diesel production rates based on your proposed engineering design.

SYSTEM
FINANCING & COMMISSIONING

ProMaxx Advanced Fuel systems are classified as a US environmental product and qualify for an attractive financing package from US EX-IM Bank.

Ex-IM offers up to 85% financing at 3.18% annual interest (or less), for loan periods of up to a maximum of 18 years. ProMaxx can direct Clients to access this financing.

When the ProMaxx system(s) is delivered to the destination site, our trained professionals will arrive to work along side Client technicians over a 2-4 weeks period to unload, install and commission the equipment. Our professionals will stay on site, training the new local crew until the unit(s) are running according to the output specifications stated in the sales contract.

SYSTEMS DEPLOYED AROUND THE GLOBE

There are currently 28 ProMaxx Advanced Fuel systems in operation around the globe, with an additional five units scheduled for manufacture in 2015.

COUNTRY	YEAR	FEEDSTOCK	UNITS
1. USA	2015*	Coal	2 units
2. Canada	2015*	Natural Gas	1 unit
3. New Zealand	2015*	Natural Gas	2 units
4. Romania	2014	Municipal solid Waste	1 unit
5. Aruba	2014	Municipal solid Waste	3 units
6. Turkey	2014	Chicken manure	1 unit
7. Colorado, USA	2013	Tires	2 units
8. Romania	2013	Tires	1 unit
9. Pennsylvania, USA	2012	Plastic	1 unit
10. Brisbane, Australia	2012	Coal	2 units
11. Seoul, S. Korea	2009	Tires	1 unit
12. California, USA	2008	Sewage Sludge	1 unit
13. China	2008	Medical Waste	1 unit
14. Tacoma, USA	2007	Tires	1 unit
15. Knoxville, USA	2007	Municipal solid Waste	1 unit
16. Milan, Italy	2006	Plastic	1 unit
17. Maine, USA	2005	Laundry waste	1 unit
18. California, USA	2004	Cow manure	1 unit
19. Taiwan	1998	Tires	3 units
20. China	1993	Tires	6 units

* *Work in progress pipeline deployments in 2015*

ProMaxx Advanced Fuels™

CONCLUSION

As the nation's fossil fuel resources gradually deplete, both government and private sector are looking for ways to cut costs, maximize efficiency and reduce reliance on expensive fuel imports. Complex logistics, uncertainty of supply and changing price structures make an already difficult situation more challenging.

New technologies such as ProMaxx's Advanced Fuel systems offer a commercially viable solution to maximize resource use efficiency and revenue, while also reducing impacts on the environment and replacing logistical uncertainty with the security of producing your own high quality fuel, 100% owned and operated.

Now is the time to invest in this new technology

SPEAK WITH A PROMAXX ADVANCED FUELS AGENT TODAY

ProMaxx Advanced Fuels™

CONTACT

GRAHAM A. HOOD
MARKETING DIRECTOR
M: [+1] (902) 825 0578
E: ghood@promaxxadvancedfuels.asia

ANDREAS DEWATMOKO
M: [+62] 856 705 6061
E: adewatmoko@promaxxadvancedfuels.asia

BENJAMIN JARVIS
M: [+62] 8118 7548 5
E: bjarvis@promaxxadvancedfuels.asia

Gedung Manggala Wanabakti, Blok IV FL.5 R. 519C
Jl. Gatot Subroto, Senayan, Jakarta Pusat 10270
T: [+62] 21 5711317
F: [+62] 21 5711317

www.promaxxadvancedfuels.asia
info@promaxxadvancedfuels.asia

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